

PURE SINE-WAVE TRANSFORMERLESS

UPS

B.E. (EL) PROJECT REPORT

Prepared by:

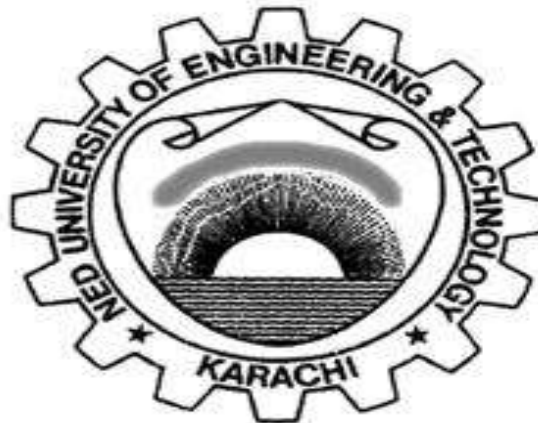
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*Through this, we would like to thanks **ALMIGHTY ALLAH**, The most merciful, for providing us the skill and strength to fulfill this task, which would not have been achieved without His blessings.*

We would like to thanks all our teachers, our Chairman and our parents for supporting us in our project.

*We would also like to thanks our Internal Advisor **Sir Muhammad Javed (Assistant Professor, NEDUET)** and our External Advisor **Mr.Muhammad Nasir (Hardware Design Engineer, Prodigy)** who guided us all the way up to the completion of this project.*

Dated: 16-10-2010

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ABSTRACT:

Electricity problem is increasing day by day in our country and unfortunately there are no signs and symptoms that it will get solve in near future. UPS which was once considered to be merely a power backup device has now became the backbone of all devices and appliances. While the position of power inverter in the market is relatively well established, there are several features that can be improved upon.

The UPS that are generally available in the market are not of good quality both in terms of wave form shape and load regulation. Majority of the available ups are square wave or at most modified sine wave. These types of UPS can't be used for the devices which are very sensitive and require pure sine wave. Those of pure sine wave that are available are not manufactured locally and are very expensive. So we are making a transformer less pure sine wave UPS which will be available at a very cheap and reasonable cost.

Its transformer less feature will make it smaller in size so that it can be placed anywhere easily. The pure sine wave output will make it a perfect choice for sensitive devices such as medical equipment, computer, server, ATM machines etc. It has a very intelligent current and voltage monitoring circuit which monitors the current and voltage and don't allow them to reach upto the level at which it could harm the load.

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SECTION 1: INTRODUCTION

SECTION: 1**UPS overview**

The primary objective of an UPS is to provide an uninterruptible power source to the connected equipment/peripherals. A normal household computer has only one source of power and in the event of any power disruption the device goes off unexpectedly may result in severe data loss, damage the computer peripherals and may cause hardware malfunctioning.

An UPS changes this scenario by providing two sources of power, primary power source and the secondary power source. The primary power source is the normal utility power source that is mounted on the wall and the battery contained secondary power source within the UPS is activated when the primary power source is disrupted. A switch controls the power sources to be used at any given time. The **Fig. 1.1** shows the basic block diagram of an UPS controlled by a switch. The switch changes from primary source

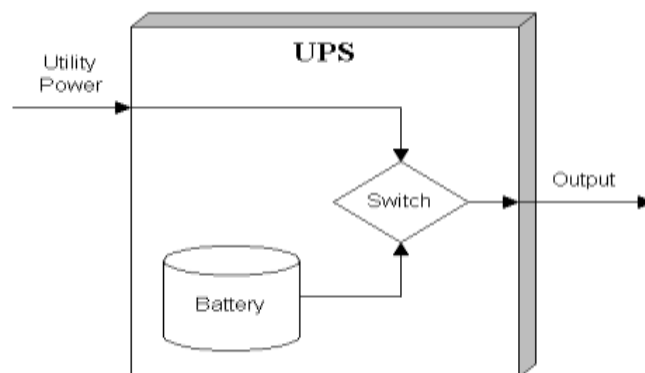


FIGURE1.1: Basic block diagram of an UPS

to secondary source automatically when it detects any utility power disruption and switches back to primary power source once the utility power source has been restored. This is termed as Transfer time or Switch time. Hold up time refers to the amount of the

SECTION: 1

time the power supply unit will continue to generate the output after utility power disruption. So in general UPS Transfer time should be compared against the Holdup time of the power supply unit before choosing a UPS, precisely the Transfer time (ms) should be lesser than the Holdup time (ms).

An Energy storing backup battery, AC-DC charger and DC-AC inverter are the components required to design an UPS.

Classification of UPS

Selecting a particular type of UPS depends upon the consumer's requirement that weighs upon the purpose of UPS, power requirement, cost, safety and durability. There are three types of UPS topologies used in the market being Standby, Ferrar-esonant standby, Online and Linear Interactive. The below section discusses about different UPS types, their basic design principles and comparison between the UPS types.

➤ Standby UPS

Standby UPS also called as Standby power supply is the simplest and cheap UPS design. In this type of UPS the primary power source is utility line power whereas the second power source is the battery. The battery charges using the utility line power whereas the battery and the

Inverter would be in the standby mode. When there is utility line power available the UPS battery and the inverter doesn't supply any power to equipment. During power disruption the transfer switch changes to the secondary power source. When utility line

SECTION: 1

power is restored the UPS switches back. They are not generally used for critical functions like important servers which is a setback for standby UPS.

➤ **Ferroresonant Standby UPS**

Ferroresonant standby UPS is an improvement design of the standby UPS. Similar to standby UPS the primary source is the utility line power and the secondary power source is the battery. The battery and inverter will be still on standby mode until the utility power is disrupted but the standby UPS's transfer switch that selects between the power sources is replaced by a ferro-resonant transformer. The secondary power source is the same as of Standby UPS.

➤ **Line-Interactive UPS**

Line interactive UPS uses a different design. Battery charger, Inverter and source selection switch of standby UPS is replaced by a combination of inverter/converter that charges the battery simultaneously. The energy is thus converted into AC for the output. Utility AC line power is still the primary power source and the battery source being secondary. During power disruption the inverter/converter discharges the battery by generating the output to the protected equipment. The inverter/converter unit is always connected to the output powering the equipment. This allows the faster transfer time during power disruption. The inverter/converter is normally embedded in the circuitry to filter out noise and spike and thus giving out a regulated output to the protected equipment. The Line-interactive UPS is commonly used in units for home and business, available in sizes up to 3,000 VA thus making it superior than the standby UPS.

SECTION: 1

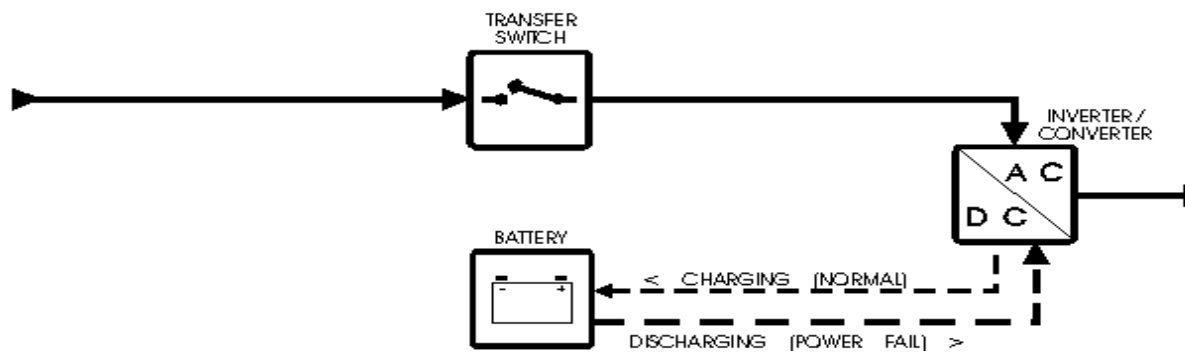


FIGURE 1.2: Block diagram of Line-Interactive UPS

➤ Online UPS

Online UPS is both similar in having transfer switch that selects between primary and secondary power sources and opposite to the standby UPS by reversing the sources. The primary source is the UPS battery and the utility line power acts as the secondary power source. Though online UPS and standby UPS have the same components inside the design is in such a way that the inverters generally would run once a month for few minute enhancing the performance and life span of the UPS. The additional engineering and the increased size and quality of components add on to make Online UPS much more expensive than the other UPS. The other disadvantage besides the cost of the online UPS is its efficiency. The power going to loads is converted from AC to DC and in turn back to AC. So the heat is dissipated in the process. Since this happens all the time the equipment consumes a lot of power. They are typically used for large servers available in sizes from about 5,000 VA up to hundreds of thousands of VA or even larger than that.

SECTION 2: THEORY

SECTION: 2

PROJECT DESCRIPTION:

We have made a High frequency Line interactive UPS. We have divided the project into following major parts:

- Battery Charger
- DC-DC converter
- Inverter
- Control circuit

BATTERY CHARGER

The following block diagram describes the fly back switch mode battery charger.

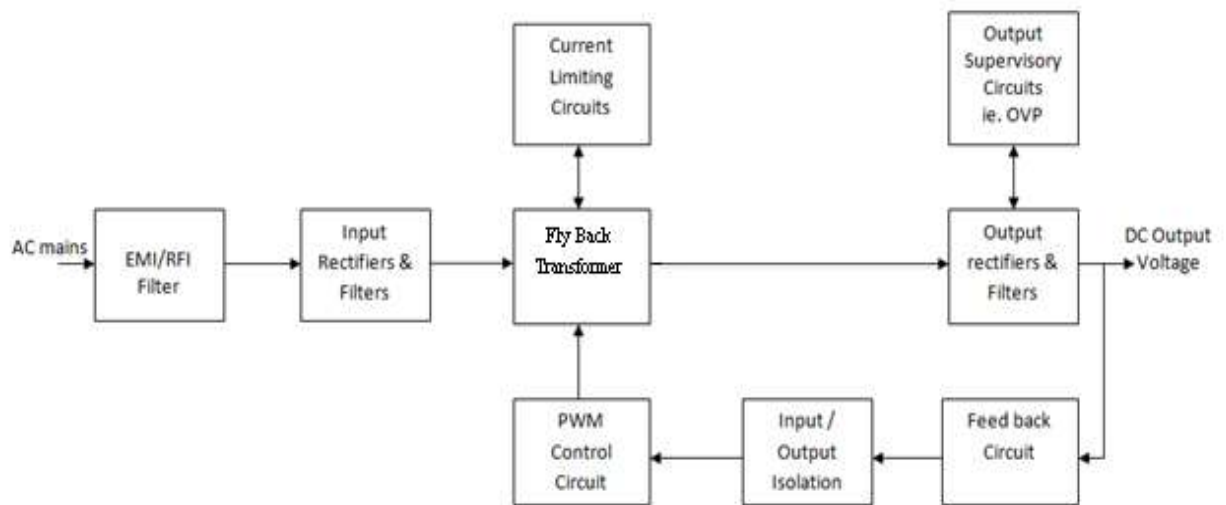


FIGURE 2.1: block diagram of fly back switch mode battery charger

EMI/RFI FILTER

EMI/RFI stands for "Electro-Magnetic Interference" and "Radio Frequency Interference".

EMI/RFI is a high frequency / low energy noise, typically of a continuous nature, that doesn't directly destroy electronic systems, but rather disrupts the proper function by

SECTION: 2

overwhelming the normal signal information within the electrical circuitry. The purpose of an EMI/RFI filter is twofold. First it stops noise from entering and disrupting the operation of the circuit. Secondly, it stops the circuit from putting EMI/RFI noise onto the power lines. The first is to protect this SMPS from malfunction or failure. The second is to protect other electrical equipment. The filter design should take care both differential-mode and common-mode noises.

INPUT RECTIFIERS & FILTERS

This stage rectifies the incoming AC directly (without any step-down), and filters out the low frequency ac component that present after the rectification. The rectifier stage employs bridge rectifiers and the filter stage employs high-grade electrolytic capacitors with high-ripple current capacity and low ESR (Equivalent Series Resistance).

FLY-BACK CONVERTER

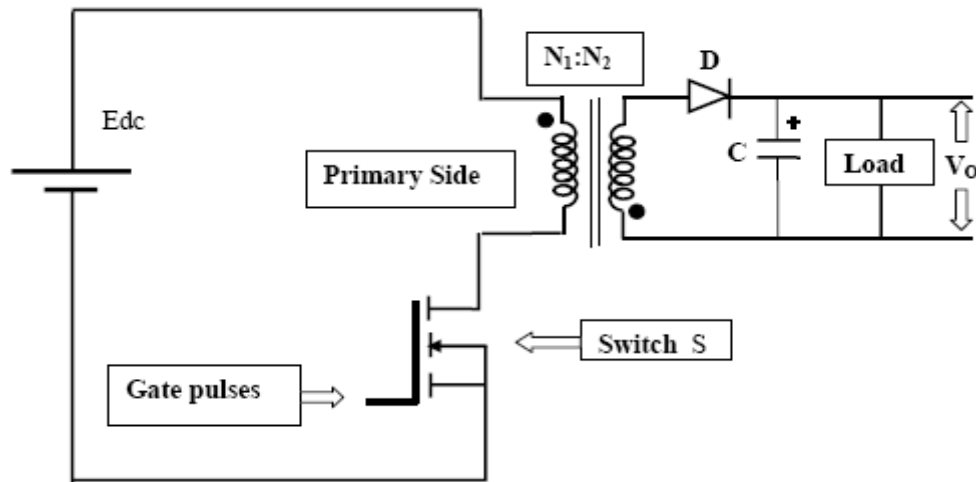
Fly-back converter is the most commonly used circuit for low output power applications where the output voltage needs to be isolated from the input main supply. The output power of fly-back type circuits may vary from few watts to less than 100 watts. The overall circuit topology of this converter is considerably simpler than other circuits. Input to the circuit is generally unregulated dc voltage obtained by rectifying the utility ac voltage followed by a simple capacitor filter. The circuit can offer single or multiple isolated output voltages and can operate over wide range of input voltage variation. In respect of energy-efficiency, fly-back power supplies are inferior to many other circuits but it's simple topology and low cost makes it popular in low output power range.

SECTION: 2

The commonly used fly-back converter requires a single controllable switch like, MOSFET and the usual switching frequency is in the range of 100 kHz. A two-switch topology exists that offers better energy efficiency and less voltage stress across the switches but costs more and the circuit complexity also increases slightly. The present lesson is limited to the study of fly-back circuit of single switch topology.

➤ Basic Topology

Fig 1 shows the basic topology of a fly-back circuit. Input to the is dc voltage derived from the utility ac supply after rectification and some filtering. The ripple in dc voltage waveform is generally of low frequency and the overall ripple voltage waveform repeats at twice the AC mains frequency. A fast switching device, like a MOSFET, is used with fast dynamic control over switch **duty ratio** (ratio of ON time to switching time-period) to maintain the desired output voltage. The transformer, in **Fig 2.2**, is used for voltage isolation as well as for better matching between input and output voltage and current requirements. Primary and secondary windings of the transformer are wound to have good coupling so that they are linked by nearly same magnetic flux. Since the primary and secondary windings of the fly-back transformer don't carry current simultaneously and in this sense **fly-back transformer works differently from a normal transformer**. In a normal transformer, under load, primary and secondary windings conduct simultaneously such that the ampere turns of primary winding is nearly balanced by the opposing ampere-turns of the secondary winding (the small difference in ampere-turns is required to establish flux in the non-ideal core).

SECTION: 2**FIGURE 2.2: Flyback Converter**

Since primary and secondary windings of the fly-back transformer don't conduct simultaneously they are more like two magnetically coupled inductors and it may be more appropriate to call the fly-back transformer as inductor-transformer. Accordingly the magnetic circuit design of a fly-back transformer is done like that for an inductor. The details of the inductor-transformer design are dealt with separately in transformer design section. The output section of the fly-back transformer, which consists of voltage rectification and filtering, is considerably simpler than in most other switched mode charger circuits. As can be seen from the circuit (**Fig. 2.2**), the secondary winding voltage is rectified and filtered using just a diode and a capacitor. Voltage across this filter capacitor is the Chargers output voltage.

OUTPUT FILTERS

The high frequency-step-down voltage from the transformer is a pulsating DC. So in order to convert it into pure DC it needs to be filtered. This stage delivers the

SECTION: 2

required regulated DC output voltage. It consists of the parallel combination of resistors and capacitors.

FEEDBACK CIRCUIT

A portion of the output voltage fed into one input of the shunt regulator which is compared against a pre-defined reference voltage V_{ref} , which is at its other input.

If there is a difference it will generate an error signal which is amplified by error amplifier. Then this error signal is sent to the PWM control circuit.

PWM CONTROL CIRCUIT

A Pulse Width Modulated signal is produced by the PWM control circuit, based on the error amplifier's output, and is fed to the base/gate of the switching element. The IC is used to generate the pulses. This PWM controls the conduction period of the switching element. If duty cycle of pulse is increased the switching element conducts for more period of time and hence greater output voltage and vice versa. So, whenever a change sensed at the output voltage will be amplified by the error amplifier, based on that PWM signal produced with more on time or off time, which will control the conduction of the switching element. If due to any reason the output voltage is decreased it is sensed by error amplifier and it provides this signal to PWM controller. The IC will increase the duty cycle of the pulse in accordance with the error signal and the output voltage will increase and come

SECTION: 2

back to the same desired level again, and thus a good regulation maintained the output.

CURRENT LIMITING CIRCUIT

Current limit circuits are basically protective networks which will limit the output current drain to a safe level; in case of an output short-circuit condition. When a short circuit or overload exists at the output of the power supply, the primary current increases dramatically, and activates the shutdown circuit. This limits the primary current to safe levels.

OUTPUT SUPERVISORY CIRCUIT

Over voltage / Current protection circuits are networks, designed to clamp the output voltage / current to a safe value. If the output voltage / current rise beyond a specified limit, this circuitry will shutdown the power supply. In our circuit we haven't made any separate special network for supervisory because we have used an intelligent IC UC3845 which is also known as current mode controller. It monitors the current and if it receives any error signal it will change the next coming pulse accordingly and hence the current will remain on the same desired level. Voltage is also maintained by the same phenomena.

SECTION: 2

DC TO DC CONVERTER

DC TO DC Converter is based on push pull topology. A push-pull topology is shown in Fig.

2.3 below consists of two transistors Q1, Q2 and a transformer T1.

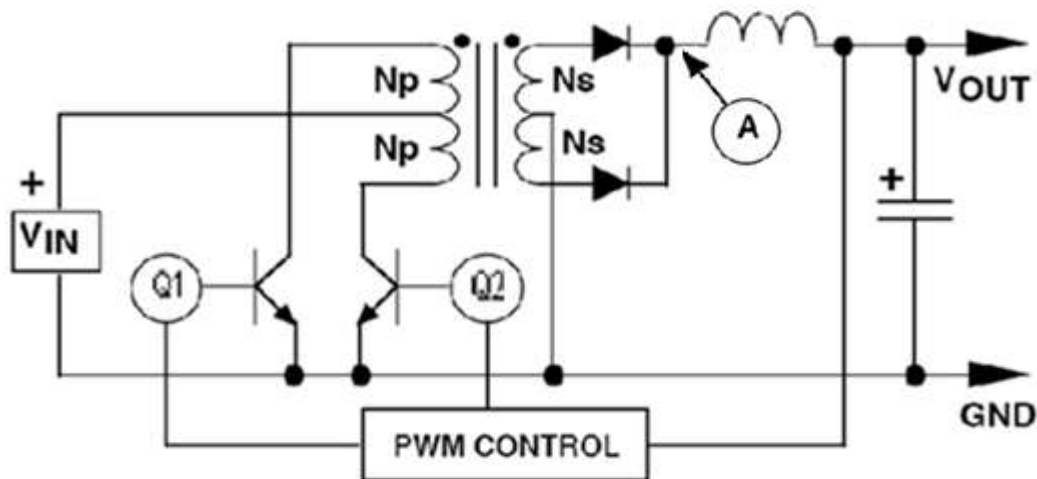


FIGURE 2.3 Typical Push Pull Converter

The control circuit provides two adjustable pulse-width, 180°-out-of-phase pulses which drive the bases of transistor Q1 and Q2. The converter operates by turning on each transistor on alternate cycles (the two Transistors are **never** on at the same time).

Transformer secondary current flows At the same time as primary current (when either of the switches is on). For example, when transistor "Q1" is turned on, the input voltage is forced across the Upper primary winding with dot-negative polarity. On the secondary side, a Dot-negative voltage will appear across the winding which turns on the bottom diode. This allows current to flow into the inductor to supply both the output capacitor and the load. When transistor "Q2" is on, the input voltage is forced across the lower primary Winding with dot-positive polarity. The same voltage polarity appears on the

SECTION: 2

secondary turns. Due to this the top diode conducts, and current flows into the output capacitor and the load.

Hence when each transistor turns "on," it applies a square-voltage pulse to its half primary of magnitude V_{dc} . On the secondary side of the transformer, there will be flat-topped square waves of amplitude $(V_{dc})(N_s/N_p) - V_d$ with a duration T_o , where V_d is an output rectifier forward drop, taken as 1 V for a conventional fast-recovery diode, and 0.5 V for a Schottky diode. The output pulses at the rectifier cathodes have a duty cycle of $2T_{on}/T$ because there are two pulses per period.

An important characteristic of a Push-Pull converter is that the switch transistors have to be able to stand off more than twice the input voltage: when one transistor is on (and the input voltage is forced across one primary winding) the same magnitude voltage is induced across the other primary winding, but it is "floating" on top of the input voltage. This puts the collector of the turned-off transistor at **twice** the input voltage with respect to ground.

The "double input voltage" rating requirement of the switch transistors means the Push-Pull converter is best suited for lower input voltage applications. It has been widely used in converters operating in 12V and 24V battery-powered systems.

SECTION: 2

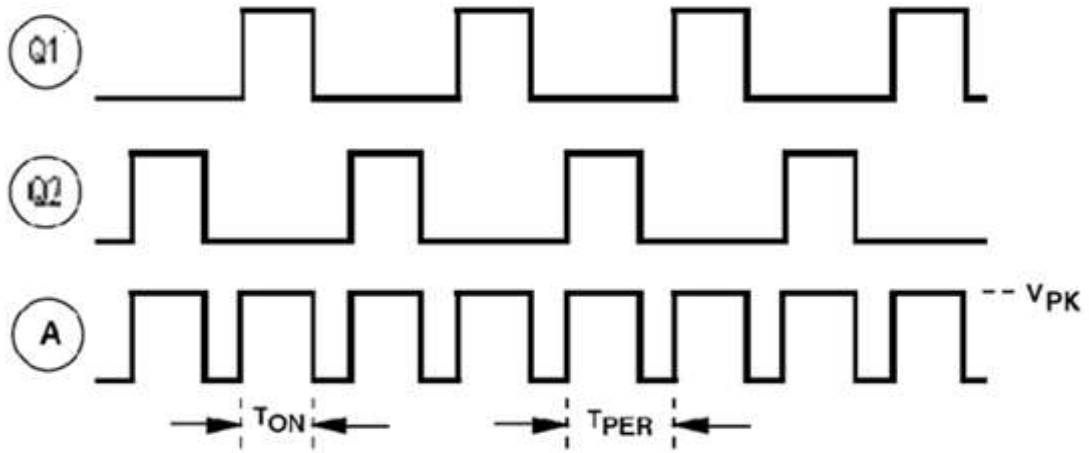


Figure 2.4 shows a timing diagram (Push Pull Converter) which details the relationship of the input and output pulses.

It is important to note that frequency of the secondary side voltage pulses is twice the frequency of operation of the PWM controller driving the two transistors. For example, if the PWM control chip was set up to operate at 50 kHz on the primary side, the frequency of the secondary pulses would be 100 kHz.

The DC output voltage is given by the equation:

$$V_m = \left[(V_{dc}) \left(\frac{N_m}{N_p} \right) - 0.5 \right] \frac{2T_{on}}{T} \quad (2.1)$$

The peak amplitude of the secondary pulses (V_{PK}) is given by:

$$V_{PK} = (V_{IN} - V_{SWITCH}) \times (N_S / N_P) - V_{RECT} \quad (2.2)$$

SECTION: 2

This highlights why the Push-Pull converter is well-suited for low voltage converters. The voltage forced across each primary winding (which provides the power for conversion) is the full input voltage minus only the saturation voltage of the switch. If MOS-FET power switches are used, the voltage drop across the switches can be made extremely small, resulting in very high utilization of the available input voltage. Another advantage of the Push-Pull converter is that it can also generate multiple output voltages (by adding more secondary windings), some of which may be negative in polarity. This allows a power supply operated from a single battery to provide all of the voltages necessary for system operation.

USING MOSFET POWER TRANSISTORS

Since most of the volt-second inequality arises from storage time inequality between the two bipolar power transistors, the problem largely disappears if MOSFETs are used, because they have no storage time. There is an added advantage, as the “on” voltage of a MOSFET transistor increases with temperature. Thus if one half primary tends to take a large current, its transistor runs somewhat warmer and its “on” voltage increases and steals voltage from the winding. This reduces the volt-second product on that side and tends to restore balance. This, of course, is qualitatively in the right direction, which is helpful but cannot be depended on to solve the flux-imbalance problem reliably at all power levels and with a worst-case combination. However, with power MOSFETs available for different types of power rating and low input voltages (as in most DC/DC converter applications), push pull converters can be and are built with a high degree of confidence.

SECTION: 2

REGULATING PULSE WIDTH MODULATOR

The Regulating pulse width IC is used for driving the gate of MOSFETS of push pull converter. It produces two square waves which are out of phase to each other on its two outputs. The frequency of the square waves can be set at the oscillator pin of IC by a parallel combination of resistor and capacitor, which will be the desired switching frequency. These pulses will drive the power MOSFETS of push pull converter. As it is the operational requirement of push pull converters that both the transistors can't be on simultaneously so the pulses produce should have a dead time i.e. both the pulse cannot be high at the same time. Since transistors take some time in transition going from on state to off state and vice versa so there should be an appropriate dead time.

OUTPUT RECTIFIER

The output of the transformer is of pulsating nature. But these pulses are of very high frequency. So the diodes used for rectification are hyper fast high frequency high voltage diodes. They convert that pulsating voltage into smooth DC voltage.

SECTION: 2

DC TO AC INVERTER

Inverter circuit is used to convert DC into AC. Inverter is built on H bridge configuration (as shown in fig below). H Bridge consists of four transistors, only two transistor works at one time. The gates of transistor are driven by PWM generator. This PWM technique is the heart of this circuit. This PWM drive of transistor is what makes this inverter a pure sine wave. Pulse width modulation (PWM) has found increasing attention in different applications in inverters for renewable energy systems, ac motor drives, and single to three-phase power conversion and telecommunication systems. Voltage source inverters using power MOSFETs and IGBTs are best suited for low to medium power applications. Many topologies of single-phase inverters have been proposed. These topologies can be classified as half-bridge or full-bridge inverter configurations. Each class can be used in two-level or multilevel switch configurations to produce an average dc voltage within each switching period which follows a fundamental sinusoidal waveform.

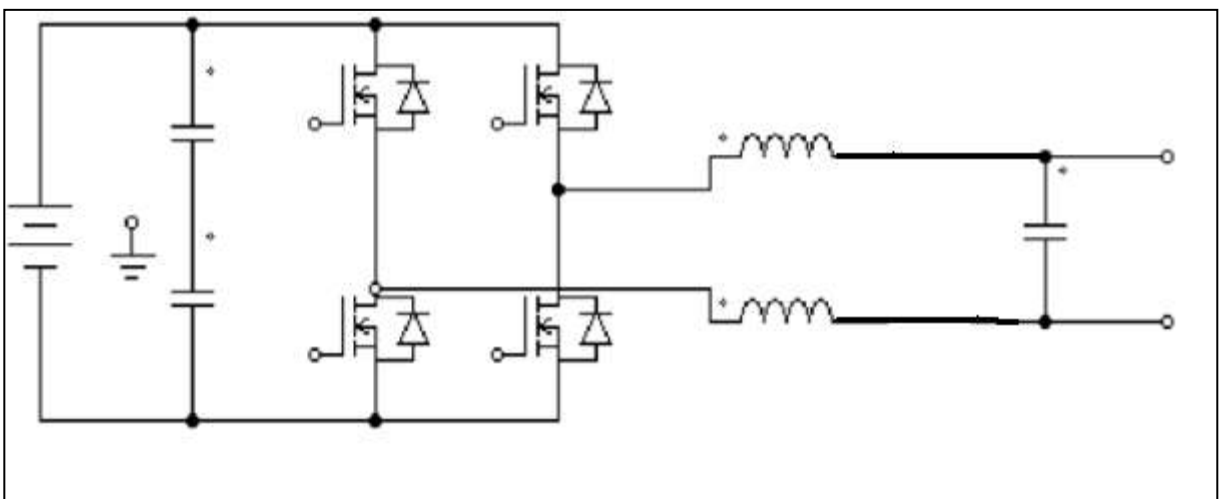


FIGURE 2.5: H-BRIDGE INVERTER

SECTION: 2

For power inverters SPWM offers greater functionality in terms of the minimization of the total harmonic distortion, reductions in size and price, and in additional inverter functional capabilities such as active filtering and reactive power support.

CONTROL CIRCUITRY

The control circuitry controls and monitors all the parameters of hardware. It drives the IC's which in turn drives the MOSFETS. The heart of control circuit is the PIC 16F877A microcontroller. Its description is given below:

- Only 35 single word instructions to learn
- All single cycle instructions except for program branches, which are two-cycle
- Operating speed: DC - 20 MHz clock input DC - 200 ns instruction cycle
- Up to 8K x 14 words of FLASH

Program Memory,

Up to 368 x 8 bytes of

Data Memory (RAM),

Up to 256 x 8 bytes of

EEPROM Data Memory

- Pinout compatible to other 28-pin

or 40/44-pin PIC16CXXX and

PIC16FXXX microcontrollers

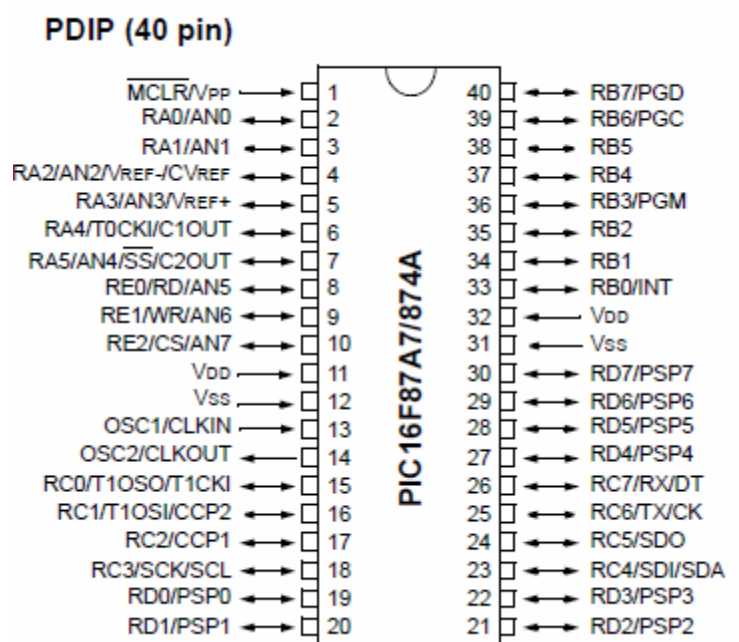
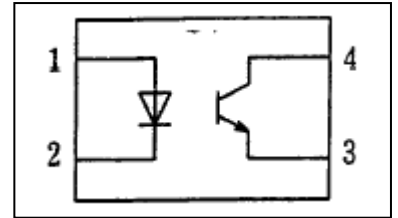


FIGURE 2.6 Pinout of PIC16F877A

SECTION: 2

MAIN SENSE:

The main sense circuit consist of a diode bridge (rectifier) and an opto-isolator to isolate mains voltage from digital circuitry. When mains supply is on it passes 5V to output which are sensed by micro-controller.



BATTERY MONITOR:

Battery monitor circuit monitors battery voltage during discharge. When battery voltage goes below 10.5V microcontroller switches off and shut-sown the inverter and other units.

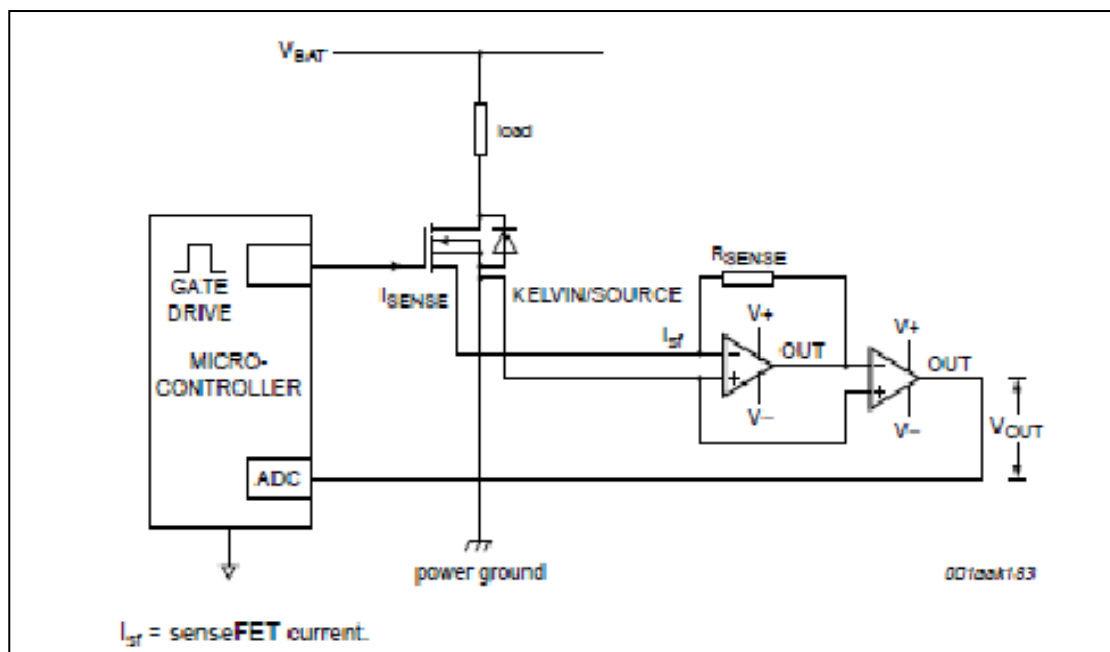


FIGURE 2.7 BATTERY MONITOR CIRCUIT

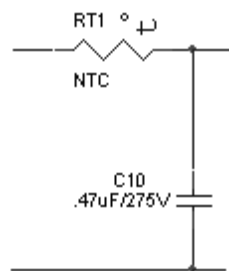
SECTION 3: HARDWARE DESCRIPTION

BATTERY CHARGER

The battery charger we have made is high frequency battery charger of 72 watts. It contains of filters, rectifier, high frequency ferrite core transformer and output control and monitor circuit. Description of each part is given below:

Filter:

The filter at the input line which filter out the line noise. It is comprised of NTC and a capacitor.



Input Rectifiers & Filters:

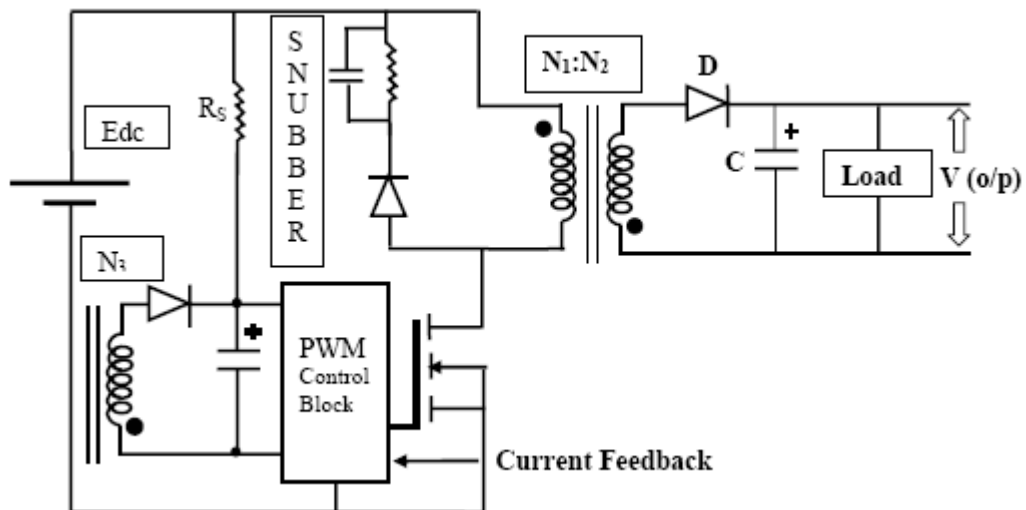
This stage rectifies the incoming AC directly (without any step-down), and filters out the low frequency ac component that present after the rectification. We have used rectifier GBU6M which steps down the AC voltage and rectifies it.

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The AC input voltage range is from 35V to 700V. It can deliver up to 6A of current. After that there is a filter circuit consisting of resistor and capacitor which filters the pulsating dc voltage and provides it to transformer.

Fly Back Converter:

This stage inverts the high DC-voltage fed from the “Input Rectifier & Filter block” to AC by chopping /switching On & Off at high frequency.



SECTION: 3

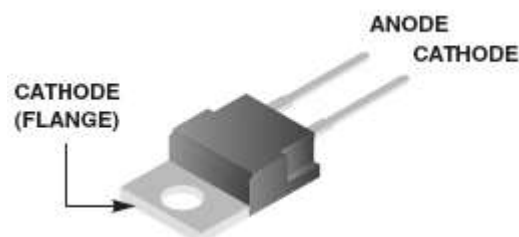
We have made the transformer on ferrite core and chose 94 kHz as our operating frequency. The turn ratio is selected in such a way that it steps down the voltage up to 12v. The switching element is connected to the PWM control circuitry which in turn is connected to feedback circuit. The detail of transformer design is given in transformer design section.

Snubber:

The snubber circuit consists of a fast recovery diode in series with a parallel combination of a snubber capacitor and a resistor. The leakage-inductance current of the primary winding finds a low impedance path through the snubber diode to the snubber capacitor. we have used diode 1N4937, resistor 15 kohm and capacitor 100nf.

Output Filters:

The high frequency-step-down voltage from the transformer is filtered at this stage. This stage delivers the required regulated DC output voltage. It consists of a diode RHRP1540 and parallel combination of resistors and capacitors which provides rectification.



SECTION: 3

RHRP1540 is a hyperfast diode i.e able to work at high frequencies. Its break down voltage is 400V and its current handling capability is up to 15 A.

Feedback Circuit:

A portion of the output voltage fed into one input of the **AS431** which is compared against a pre-defined reference voltage V_{ref} , which is at its other input. Below is shown AS431



The AS431 is a three-terminal adjustable shunt regulator providing a highly accurate bandgap reference. The adjustable shunt regulator is ideal for a wide variety of linear applications that can be implemented using external components to obtain adjustable currents and voltages. In the standard shunt configuration, the combination of low temperature coefficient (TC), sharp turn-on characteristics, low output impedance and programmable output voltage make this precision reference a perfect zener diode replacement. It provides the reference signal and continuously compares the output signal with its reference. If there is a difference it will generate an error signal which is amplified by error amplifier. Then this error

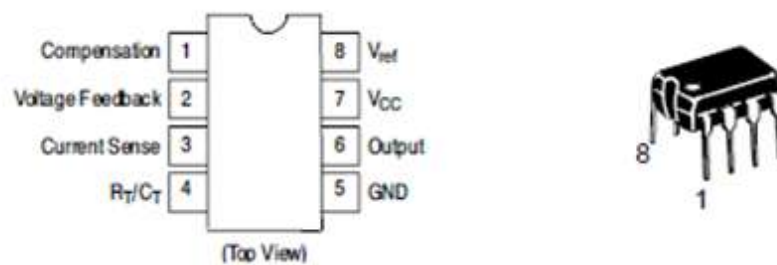
SECTION: 3

signal is sent to the PWM control circuit which by changing the operating frequency of transistor changes the output voltage.

PWM Control Circuit:

A Pulse Width Modulated signal is produced by the PWM control circuit, based on the error amplifier's output, and is fed to the base/gate of the switching element.

The current mode PWM controller ic **UC3845** is used to generate the pulses.



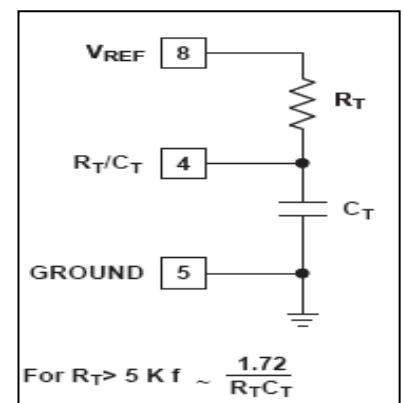
PIN FUNCTION DESCRIPTION

PIN no.	FUNCTION	DESCRIPTION
1	Compensation	This pin is Error Amplifier output and is made available for loop compensation.
2	Voltage Feedback	This is the inverting input of the Error Amplifier. It is normally connected to the switching power supply output through a resistor divider.
3	Current Sense	A voltage proportional to inductor current is connected to this input. The PWM uses this information to terminate the output switch conduction.
4	RT/CT	Oscillator frequency and maximum Output duty cycle are programmed by connecting resistor RT to V _{ref} and capacitor CT to ground. Operation to 1.0 MHz is possible.
5	GND	This pin is combined control circuitry and power ground

SECTION: 3

6	Output	This output directly drives the gate of a power MOSFET. Peak currents up to 1.0 A are sourced and sunk by this pin. The output switches at one-half the oscillator frequency.
7	VCC	This pin is the positive supply of the control IC.
8	Vref	This is the reference output. It provides charging current for capacitor CT through resistor RT.

The oscillator frequency is programmed by the values selected for the timing components R_T and C_T . Capacitor C_T is charged from the 5.0 V reference through resistor R_T to approximately 2.8 V and discharged to 1.2 V by an internal current sink.



This PWM controls the conduction period of the switching element. If duty cycle of pulse is increased the switching element conducts for more period of time and hence greater output voltage and vice versa. So, whenever a change sensed at the output voltage will be amplified by the error amplifier, based on that PWM signal produced with more on time or off time, which will control the conduction of the switching element. If due to any reason the output voltage is decreased it is sensed by error amplifier and it provides this signal to uc3845. The IC will increase the duty cycle of the pulse in accordance with the error signal and the output voltage will increase and come back to the same desired level again, and thus a good regulation maintained the output.

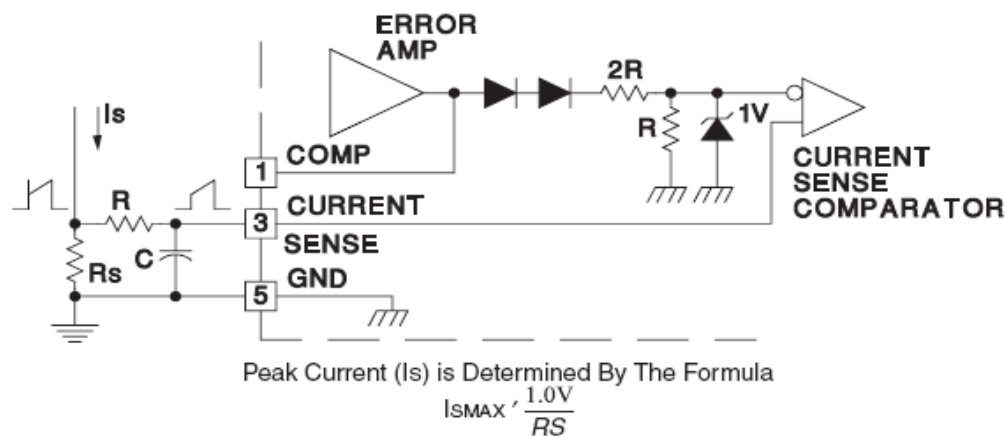
SECTION: 3

Current Limiting Circuits :

In order to keep the current within safe levels, IC uc3845 has the current sensing pin, shown in fig below which senses the current and maintains the pulses duty cycle accordingly so that the current remains in the safe level.

Output Supervisory Circuit:

Over voltage / Current protection circuits are networks, designed to clamp the output voltage / current to a safe value. If the output voltage / current rises beyond a specified limit, this circuitry will shutdown the power supply. In our circuit we haven't made any separate special network for supervisory because we have used an intelligent IC UC3845 which is also known as current mode controller. It monitors the current (see fig below) and if it receives any error signal it will change the next coming pulse accordingly and hence the current will remain on the same desired level. Voltage is also maintained by the same phenomena.



DC TO DC CONVERTER

The function of DC to DC converter is to take the 12 volt from the battery and step it up to 300volts. Then this voltage is feed in to the inverter circuit which converts it into AC sinusoidal wave. As discussed in theory section that the DC to DC converter is based on the push pull topology. It consists of PWM regulating IC, transformer, switching transistors (MOSFETS) and output rectifier.

TRANSFORMER

The center part of this circuit is the high frequency transformer. It is center tapped transformer with its center leg connected to battery. The other two legs are connected to switching transistors. There is a constant battery voltage at the center of transformer. But the ground shifts up and down as the transistors switch to ground alternatively. So we have to select the core which matches our output voltage and switching frequency requirement plus it should be small in size. The detail; construction of transformer is given in transformer section.

SWITCHING MOSFETS

Once the transformer is made the other most important decision is to select the transistor. Since the input side or the primary side of transformer where transistor would be connected is subjected to very high current so therefore we have to use power MOSFETS. The transistor that we are using is **IRF 3250** shown below:

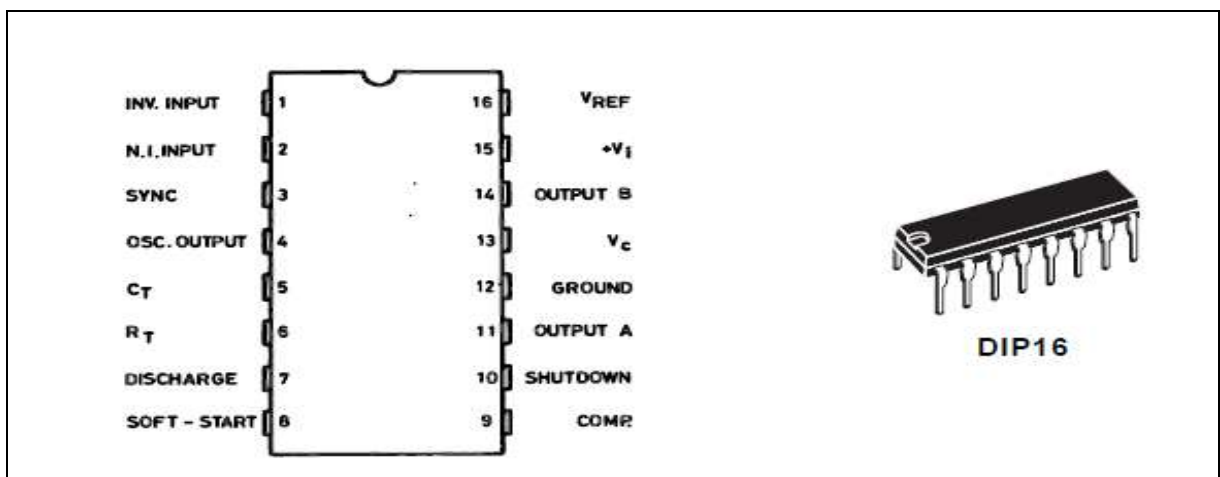
SECTION: 3



It has extremely low on-resistance per silicon area. This benefit, combined with the fast switching speed made it very reliable for the use in this type of applications. It can carry up to 110 A with a V_{DSS} of 55V. A heat sink should be attached while operating otherwise due to high current transistor get so much heat that it can get damaged.

PULSE WIDTH MODULATOR IC

The IC used for pulse width modulation is **SG 3525** shown below:



The SG3525A series of pulse width modulator integrated circuits are designed to offer improved performance and lowered external parts count when used in designing all types of switching power supplies. The on-chip + 5.1 V reference is trimmed to $\pm 1\%$ and

SECTION: 3

the input common-mode range of the error amplifier includes the reference voltage eliminating external resistors.

Its oscillation frequency is set by connecting a RC network between pin 5 and pin6. This oscillator network will set the output pulse frequency, which in turn switches the MOSFETS. The value of resistor and capacitor is set by following formula:

$$f = 1 / \{CT (0.7 RT + 3 RD)\}$$

So we have selected $RT=15\text{kohm}$, $CT=1\text{nf}$ and $RD=220\text{ ohm}$ to select 90 kHz switching frequency.

It has two output pins one at pin 11 and other at pin 14. The output pulses at both the output pins are 180 degree out of phase hence enabling one transistor at a time. But there is always a danger due to some delays that two transistor might conduct simultaneously. If this happens transistors will get blown up. To avoid this condition SG3525 has built in dead time. It automatically takes care of this very serious problem.

In our circuit we are operating this IC on 45% duty cycle. It has this feature that it compares the output signal with a reference voltage and if it sees any error it changes the duty cycle so as to maintain the output voltage.

OUTPUT RECTIFIER

The output of the transformer is pulsating in nature but we have to apply a constant DC level to inverter. For that we have to rectify the pulsating DC and make it into constant

SECTION: 3

dc. For that we have to use diode but we cannot use normal diode because the output frequency is very high. For that we will use hyperfast high frequency diodes. The diodes we used are FES8GT.



These diodes can carry current up to 8 Amperes and their breakdown voltage is 400 V.

INVERTER

The inverter converts DC voltage that comes from DC to DC converter into 220 V AC. It consists of SPWM driver, opto coupler and H Bridge transistors. Brief description of all these are given below:

SPWM DRIVER:

PWM driver circuitry consists of 74HC257 (Quad-2 Input Data Selector\Multiplexer).

The IC selects a (4-bit) nibble from either the A or B inputs as determined by the Select input. The nibble is presented at the outputs in non-inverted form when the Output Enable pin is at a low level. A high level on the Output Enable pin switches the outputs into the high-impedance state.

SECTION: 3**Function Table**

Inputs				Outputs
Output Control	SELECT	A	B	Y
L	L	L	X	L
L	L	H	X	H
L	H	X	L	L
L	H	X	H	H
H	X	X	X	High impedance

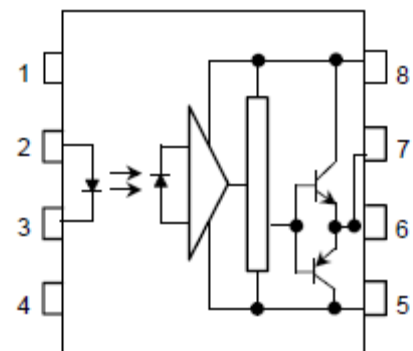
Features:

- Output Drive Capability: 15 LSTTL Loads
- Outputs Directly Interface to CMOS, NMOS, and TTL
- Operating Voltage Range: 2 to 6 V
- Low Input Current: 1 μ A
- High Noise Immunity Characteristic of CMOS Devices

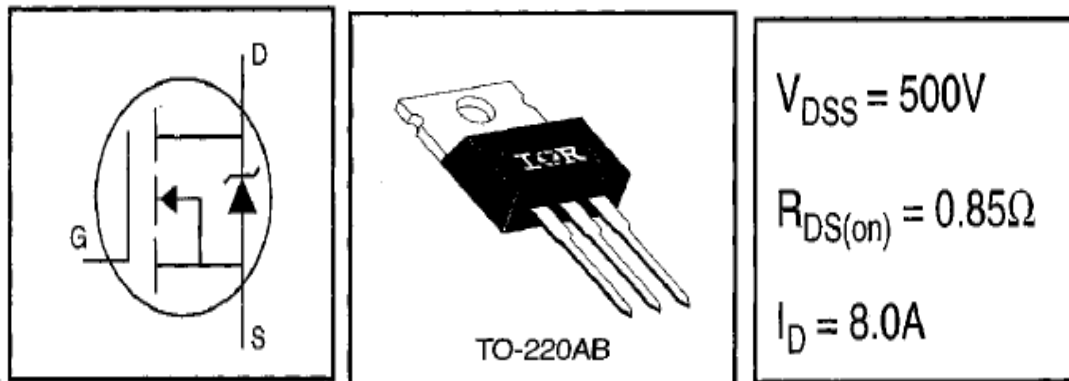
TLP250 opto-coupler:

They are used in the Power MOSFET's gate drives.

- 1 : N.C.
- 2 : Anode
- 3 : Cathode
- 4 : N.C.
- 5 : GND
- 6 : VO (Output)
- 7 : VO
- 8 : VCC

Pin Configuration (top view)**H-BRIDGE MOSFETS (IRF840)**

These are the MOSFETS that are used in Full bridge. They can carry up to 8 A and stand upto 500 V. They are driven by the SPWM technique in such a way that they generate pulses which are filtered into a sine wave.

SECTION: 3**FILTER:**

After getting the pulses out from full bridge, it is this filter that constructs a sine wave output. The filter we use is a simple LC filter. Its cutoff frequency should be 10 times lower than the carrier frequency of PWM wave and 10 times higher than the desired output frequency so our carrier frequency is 4 KHZ and output frequency is 50 HZ. So our filter's cutoff frequency range lies between 400 HZ-500 HZ.

TRANSFORMER DESIGN

The main component in the push pull converter is the transformer that step ups the 12v of battery into 320v. But this is not the normal core transformer. For these types application ferromagnetic core transformers are used. Ferrites are ceramic ferromagnetic materials having a crystalline structure consisting of mixtures of iron oxide with either manganese or zinc oxide. Their eddy current losses are negligible, as their electrical resistivity is very high. Core losses comprise mainly hysteresis losses which are low enough to permit use of some materials up to a frequency of 1 MHz. This high frequency operation reduced the size of the core required for desired output power and hence

SECTION: 3

reduced the size of overall transformer multiple times when compared with iron core transformer for same power.

The design of a transformer starts with the initial selection of a core to satisfy the desired total output power. The available output power from a particular core depends on the operating frequency, the operating flux density swings, the core's area A_e , the bobbin winding window area A_b , and the current density in each winding. Decisions on each of these parameters are interrelated, and choices are made to minimize the transformer size and its temperature rise

Energy Storage in a Transformer

Ideally, a transformer stores no energy—all energy is transferred instantaneously from input to output. In practice, all transformers do store some undesired energy:

- Leakage inductance represents energy stored in the non-magnetic regions between windings, caused by imperfect flux coupling. In the equivalent electrical circuit, leakage inductance is in series with the windings, and the stored energy is proportional to load current squared.
- Mutual inductance (magnetizing inductance) represents energy stored in the finite permeability of the magnetic core and in small gaps where the core halves come together. In the equivalent circuit, mutual inductance appears in parallel with the windings. The energy stored is a function of the volt-seconds per turn applied to the windings and is independent of load current.

PUSH PULL CONVERTER TRANSFORMER DESIGN

Maximum Flux Change (Flux Density Swing) Selection

From Eq. 3.1, it is seen that the number of primary turns is inversely proportional to dB , the flux swing. It would seem desirable to maximize dB so as to minimize N_p since fewer turns would mean that a larger wire size could be used, resulting in higher permissible currents and more output from a given core. Also, fewer turns would result in a less expensive transformer and lower stray parasitic capacities. From the hysteresis loop analysis, it is desirable to stay below ± 2000 G, where the magnetizing current starts increasing rapidly. So initially a good choice would appear to be ± 2000 G (0.2 tesla). But we must also consider core losses. Ferrite core losses increase at about the 2.7th power of the peak flux density and at about the 1.6th power of the operating frequency. Hence, up to about 50 kHz, core losses do not prohibit operation to ± 2000 G, and it may appear desirable to operate at that flux level. However, to prevent core saturation under transient conditions, it is better to provide a wider margin. Faraday's law solved for the flux change

$$dB = \frac{(V_{dc})(T_{on}) \times 10^8}{N_p A_e} \quad 3.1$$

Above 50 kHz, the core losses increase rapidly and force a lower flux density selection. At 100 to 200 kHz, the peak flux density may be limited to 1200 to achieve an acceptably low core temperature rise.

Maximum Power Transistor On-Time Selection

Equation 3.1 has shown that the converter keeps the output voltage V_m constant by increasing T_{on} as V_{dc} decreases. Thus the maximum "on" time T_{on} occurs at the

SECTION: 3

minimum specified DC input voltage V_{dc} . But in this type of converter the maximum “on” time must not exceed half the switching period T . If it were to do so, the reset volt-second product would be less than the set volt-second product, and after a very few cycles, the core would drift into saturation and destroy a power transistor. Moreover, because of the inevitable storage time in bipolar transistors, the base drive “on” time cannot be as large as a full half period, as the storage time would cause an overlap with the opposite transistor. This would result in immediate failure, because the two power transistors would effectively short out the winding. Each transistor would take large currents at the full supply voltage and would rapidly be destroyed. Thus, to ensure that the core will always be reset within one period and eliminate any possibility of simultaneous conduction, whenever the DC input voltage is at its minimum V_{dc} and the feedback loop is trying to increase T_{on} to maintain V_m constant, the maximum “on” time will be constrained by some kind of a clamp so as to never be more than 80% of a half period. Then in Eq. 3.1, for the specified V_{dc} , T and for $T_{on} = 0.8T/2$, the ratio N_m/N_p will be fixed to yield the desired output V_m .

Primary Turns Selection

The number of primary turns is determined by Faraday’s law equation. From it N_p is fixed by the minimum voltage across the primary (V_{dc}) and the maximum “on” time, which, as earlier, is to be no more than $0.8T/2$. Then

$$N_p = \frac{(V_{dc})(0.8T/2) \times 10^8}{A_e dB} \quad (3.2)$$

SECTION: 3

Since A_e in Eq. 3.2 is fixed by the selected core, V_{dc} and T are specified and the number of primary turns is fixed as soon as dB (the desired flux change in $0.8T/2$) is decided on.

This decision is made as follows.

Secondary Turns Selection

The turns for the main outputs are calculated from Eq. 3.3 in accordance with the specified, or calculated, voltage requirements.

$$V_s = \left[(V_{dc}) \frac{N_s}{N_p} - 1 \right] \frac{2T_{on}}{T} \quad (3.3)$$

We see that the input voltage V_{dc} and T have been specified. The maximum “on” time T_{on} has been arbitrarily set at $0.8T/2$, and N_p has been calculated from Faraday’s law (see Eq. 3.2) for the known A_e for the selected core. Flux swing dB has been set at 3200G for frequencies under 50 kHz and to minimize core losses. Lower values will be used at higher frequencies as discussed earlier.

Primary rms Current Calculation and Wire Size Selection

Each half primary carries only one of the I_{pft} pulses per period, and hence its duty cycle is $(0.8T/2)/T$ or 0.4.

$$I_{rms} = 0.632 \frac{1.56 P_o}{\underline{V_{dc}}} = \frac{0.986 P_o}{\underline{V_{dc}}} \quad (3.4)$$

SECTION: 3

This gives the rms current in each half primary in terms of the known parameters: output power and the specified minimum DC input voltage. A conservative practice in transformer design is to operate the windings at a current density of 500 circular mils per rms ampere. There is nothing absolute about this; current densities of 300 circular mils per rms ampere are frequently used for windings with only a few turns. As a general rule, however, densities greater than 300 circular mils per rms ampere should be avoided, as that will cause excessive copper losses and temperature rise.

Thus at 500 circular mils per rms ampere, the required number of circular mils for the half primaries is Circular mils = 500

$$\begin{aligned}\text{Circular mils} &= 500 \frac{0.986 P_o}{V_{dc}} \\ &= 493 \frac{P_o}{V_{dc}}\end{aligned}\quad (3.5)$$

Notice that this is also in terms of known values—output power and specified minimum DC input voltage. Proper wire size can then be chosen from wire tables at the circular mils given by Eq. 3.5.

Secondary Peak, rms Current, and Wire Size Calculation

The rms current in each half secondary is

$$I_{s(\text{rms})} = I_{dc} \sqrt{D} = I_{dc} \sqrt{0.4} = 0.632 I_{dc} \quad (3.6)$$

SECTION: 3

At 500 circular mils per rms ampere, the required number of circular mils for each half secondary is

$$\begin{aligned}\text{Secondary circular mil requirement} &= 500(0.632)I_{dc} \\ &= 3.16I_{dc}\end{aligned}\quad (3.7)$$

Transistor Voltage Stress and Leakage Inductance Spikes

In push pull configuration when either transistor is “on,” the opposite transistor’s collector is subject to at least twice the DC supply voltage, since both half primaries have an equal number of turns and are in series, with the center tap connected to the supply. However, the maximum stress is somewhat more than twice the input voltage. An additional contribution comes from the so-called leakage inductance spikes. These come about because there is effectively a small inductance (leakage inductance L_l) in series with each half primary. At the instant of turn “off,” current in the transistor falls rapidly at a rate di/dt , causing a positive-going spike of amplitude $e = L_l di/dt$ at the bottom end of the leakage inductance. Conservative design practice assumes the leakage inductance spike may increase the stress voltage by as much as 30%, more than twice the maximum DC input voltage. Hence the transistors should be chosen so that they can tolerate with some safety margin a maximum voltage stress V_p of

$$V_p = 1.3(2V_{dc}) \quad (3.8)$$

SECTION: 3

Leakage inductance delays the transfer of current between switches and rectifiers during switching transitions. These delays, proportional to load current, are the main cause of regulation and cross regulation problems. Mutual inductance and leakage inductance energy causes voltage spikes during switching transitions resulting in EMI and damage or destruction of switches and rectifiers. The magnitude of the leakage inductance is not easily calculable. It can be minimized by use of a transformer core with a long center leg and by sandwiching the secondary windings (especially the higher current ones) in between halves of the primary. A good transformer should have leakage inductance of no more than 4% of its magnetizing inductance. Leakage inductance spikes can be minimized by addition of a snubber circuit (a capacitor, resistor, and diode combination) connected to the transistor collector as shown in Figure 3.1 *below*. The stored energy then ends up as loss in the snubbers or clamps.

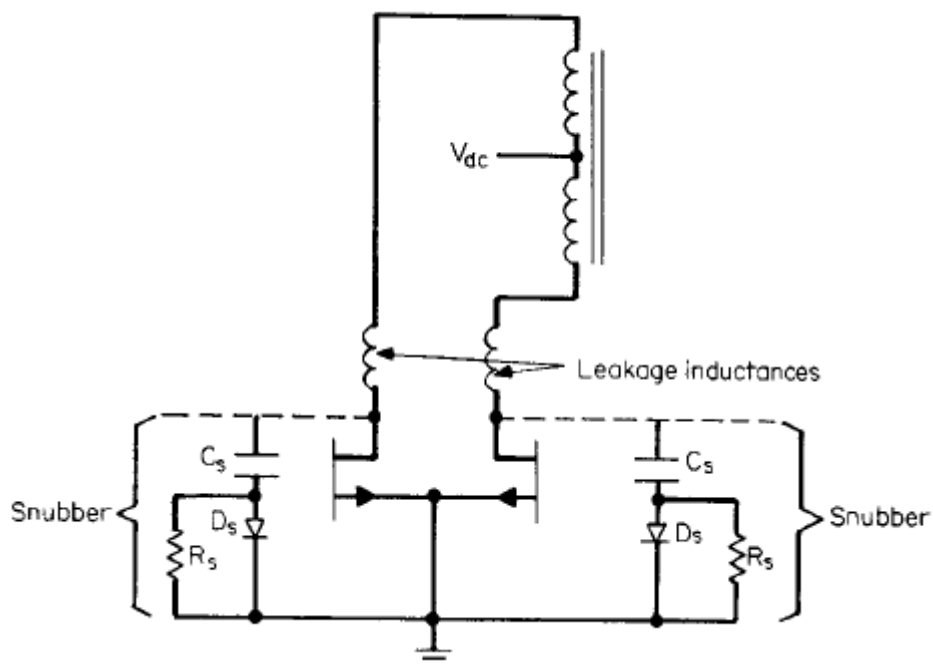


FIGURE3.1: Push Pull converter with Snubber circuit

SECTION: 3**Balancing Core and Winding Losses**

At higher operating frequencies, when the core is usually loss-limited, not saturation limited, total losses are at a broad minimum when core losses are approximately equal to or a little less than winding losses. Likewise, winding losses are at a minimum and well distributed by making the rms current density approximately equal in all windings. With a bridge or half-bridge primary, which has good winding utilization, and center-tapped secondary's which have poor utilization, rms current densities will be approximately equalized when the primary conductor cross-section area is 40% and the secondary's 60% of the available area. In most other cases, primary and secondary conductor areas should be 50%/50%, including: The above allocations can be impossible to achieve because the number of turns in each winding must be an integral number. In a low voltage secondary, 1.5 turns may be required for optimum balance between core and winding losses. With one turn, the flux swing and core loss may be much too large; with two turns the winding loss becomes too great. At either extreme, it may be impossible to meet temperature rise or absolute loss limits. A larger core may be required to resolve this problem.

OUR DESIGN:

For our converter ratings:

$V_{dc}(\max)=13$

$V_{dc}(\min)=10.5$

$F=90\text{KHZ}$

$T=11.11\text{ }\mu\text{ sec}$

Duty cycle=0.45

$V_o=320\text{V}$

$P_o=1\text{KVA}$

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We selected the core ETD 49 due to high power requirement. By using above mention formulas our design parameters comes out to be

$N_p=2$ (1 turn each primary) with AWG=4

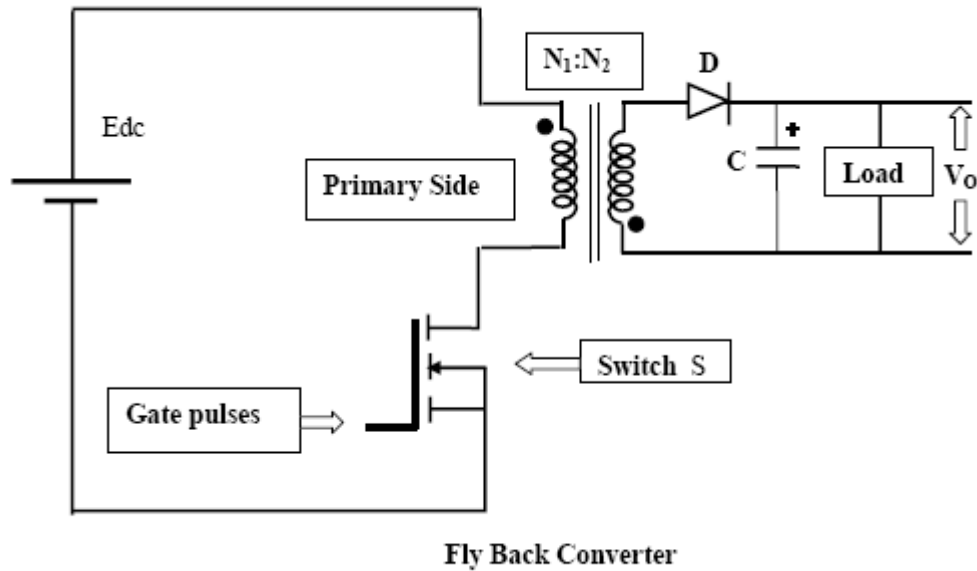
Since it is practically impossible to wound AWG No.4 so we used paralleling technique i.e. we parallel 10 AWG 13 wires.

$N_s=33$ with AWG=14

SECTION: 3

FLYBACK TRANSFORMER DESIGN

The fly back transformer is shown below:



The transformer input voltage from rectifier is 315 volts. So in order to get the output of charger 12V we have to design the transformer which fulfills these requirements. We will make it on ferrite core and will use it at very high frequency in order to reduce the weight and size of transformer. Therefore we have selected the core ETD44 which meet our requirement and constraints.

Establish the Primary/Secondary Turns Ratio

For the most expedient design, there are a number of decisions that should be made in the following logical sequence. First select a core size to meet the power requirements. Next choose the primary/master secondary turns ratio N_p/N_{sm} to determine the maximum “off”-voltage stress V_{ms} on the power transistor in the absence of a leakage

SECTION 3

inductance spike as follows: Neglecting the leakage spike, the maximum transistor voltage stress at maximum DC input V_{dc} and for a 1-V rectifier drop is

$$\overline{V_{ms}} = \overline{V_{dc}} + \frac{N_p}{N_{sm}}(V_o + 1) \quad (3.9)$$

where V_{ms} is chosen sufficiently low so that a leakage inductance spike of $0.3V_{dc}$ on top of that still leaves a safety margin of about 30% below the maximum pertinent transistor rating (V_{ds}).

Ensure the Core Does Not Saturate

To ensure that the core does not drift up or down its hysteresis loop, the condition given below should be fulfilled. Assume that the “on” drop of Q_1 and the forward drop of the rectifier D_2 are both 1 V:

$$(\overline{V_{dc}} - 1)\overline{T_{on}} = (V_o + 1)\frac{N_p}{N_{sm}}T_r \quad (3.10)$$

where T_r is the reset time required for the secondary current to return to zero. $\overline{T_{on}}$ is the maximum “on” time, which occurs when V_{dc} is a minimum, plus the reset time T_r is only 80% of a full period.

Adjust the Primary Inductance

The primary inductance of the transformer which is the center of this flyback topology can be calculated from:

$$L_p = \frac{(\overline{V_{dc}}\overline{T_{on}})^2}{2.5T P_o} \quad (3.11)$$

SECTION: 3

Check Transistor Peak Current

If the transistor is a bipolar type, it must have an acceptably high gain at the peak current operating current I_p . This is

$$I_p = \frac{V_{dc} \overline{T_{on}}}{L_p} \quad (3.12)$$

where V_{dc} is specified and T_{on} is known from the duty cycle we use and L_p is calculated from Eq. 5.3. If Q_1 is a MOSFET, it should have a peak current rating about 5 to 10 times the value calculated from Eq. 5.4 so that its “on”-state resistance is low enough to yield an acceptably low voltage drop and power loss.

Primary RMS Current

The primary current is a peak amplitude I_p (Eq.3.12) at a maximum duration T_{on} out of every period T . Its RMS value is

$$I_{rms(primary)} = \frac{I_p}{\sqrt{3}} \sqrt{\frac{T_{on}}{T}} \quad (3.13)$$

where I_p is given by Eq. 3.12 .Once the current is known wire size can be selected.

Secondary RMS Current

The secondary current is a of peak amplitude $I_s = I_p(N_p/N_s)$ and duration T_r . Primary/secondary turns ratio N_p/N_s is given by Eq. 3.9 and $T_r = (T - T_{on})$. Secondary RMS current is then

$$I_{rms(secondary)} = \frac{I_p(N_p/N_s)}{\sqrt{3}} \sqrt{\frac{T_r}{T}} \quad (3.14)$$

SECTION: 3**OUR DESIGN:**

By taking N_p/N_s and $T_{on}=0.45$ and putting all values in the above mention formulas. We came up with the following results

$N_p=55$ with AWG=21

$N_s=3$ with AWG=10

Since AWG 10 is very hard to wound, we use wire paralleling technique .we took two AWG 15 wire made it parallel and wound it.

And core is ETD44.

SECTION 4: SOFTWARE DESCRIPTION

SECTION: 4

We start our discussion of software with sine wave creation part but in order to discuss we first need to understand what type of signal are required to be generated from Micro-controller and why. Let us have a quick review of how a dc signal can be converted to AC signal using Pulse width modulation. The basic idea is to change the normal dc reference signal in PWM by Reference sine wave as explain in following discussion.

PULSE WIDTH MODULATION (PWM):

In electronic power converters and motors, PWM is used extensively as a means of powering alternating current (AC) devices with an available direct current (DC) source or for advanced DC/AC conversion. Usually the output of normal PWM employed to run motor gives output in the form of Square wave rather than sine wave as shown in Figure 4.1. The speed of the motor is control by varying the duty cycle of the output wave which control the average voltage supplied to load.

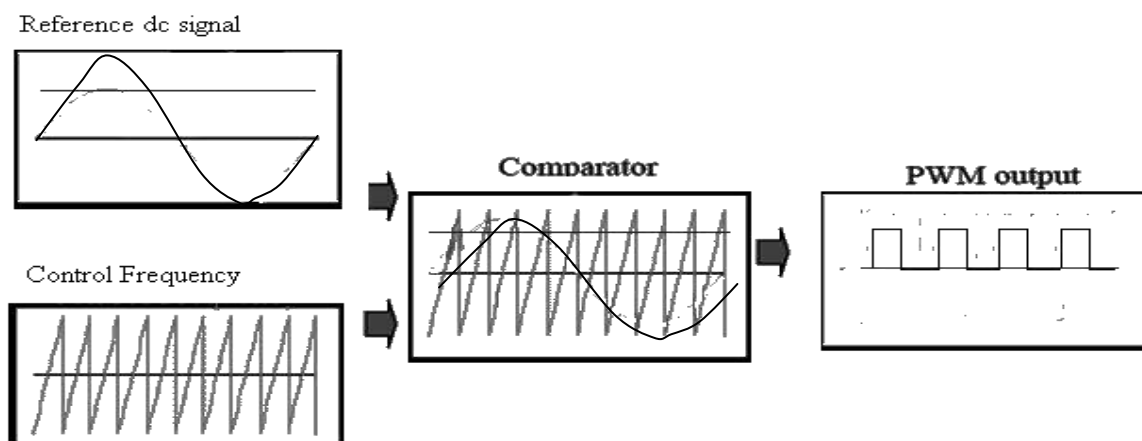


Figure 4.1

SINOSOIDAL PULSE WIDTH MODULATION:

Variation of duty cycle in the PWM signal to provide a DC voltage across the load in a specific pattern will appear to the load as an AC signal, or can control the speed of

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motors that would otherwise run only at full speed or off. This is further explained in later section. For sine wave creation at output reference signal also known as carrier signal is sinusoidal as shown is Figure 4.2.

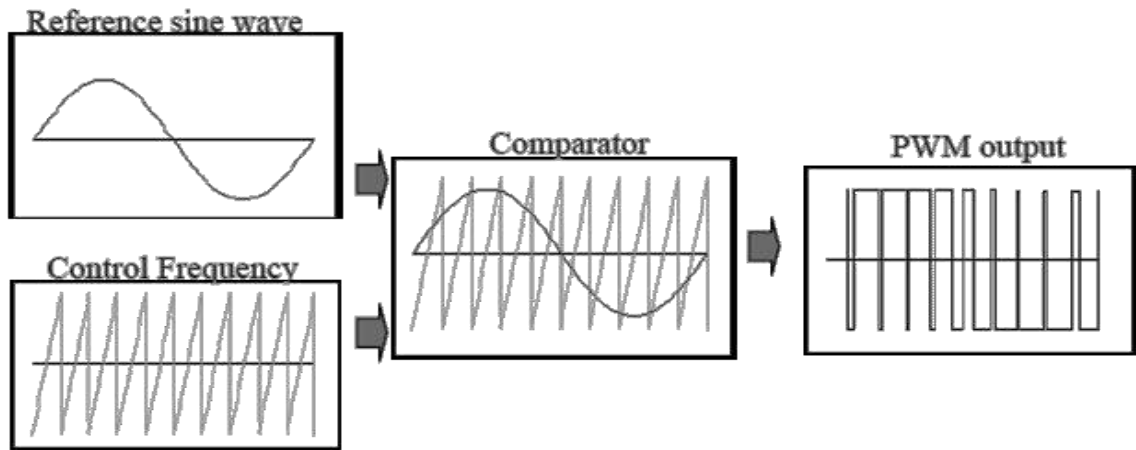


Figure 4.2

TYPES OF PULSE WIDTH MODULATION:

There are two ways of creating PWM depending upon the either the wave is generated for half cycle (wave) or full cycle (wave). The method for these two are named as

- 1- Bi-polar PWM
- 2- Uni-polar PWM

1. Bi-polar PWM

Bi-polar PWM refers to the technique of switching a full-bridge inverter in

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complementary pairs so that only two possible output voltages are available: V and $-V$.

Bipolar PWM output contains either $+V$ or $-V$ always. Hence, each state change involves a transition by $2V$. Hence switching harmonic content in this scheme is more, in addition, the switching harmonic content is the highest when the PWM is trying to generate zero volt output after the averaging filter. Hence, it is difficult to synthesize good 'zero crossings' in a bipolar scheme unless heavy filtering (with consequent degradation in response time) is used. Figure 4.3 shows how bipolar PWM is generated.

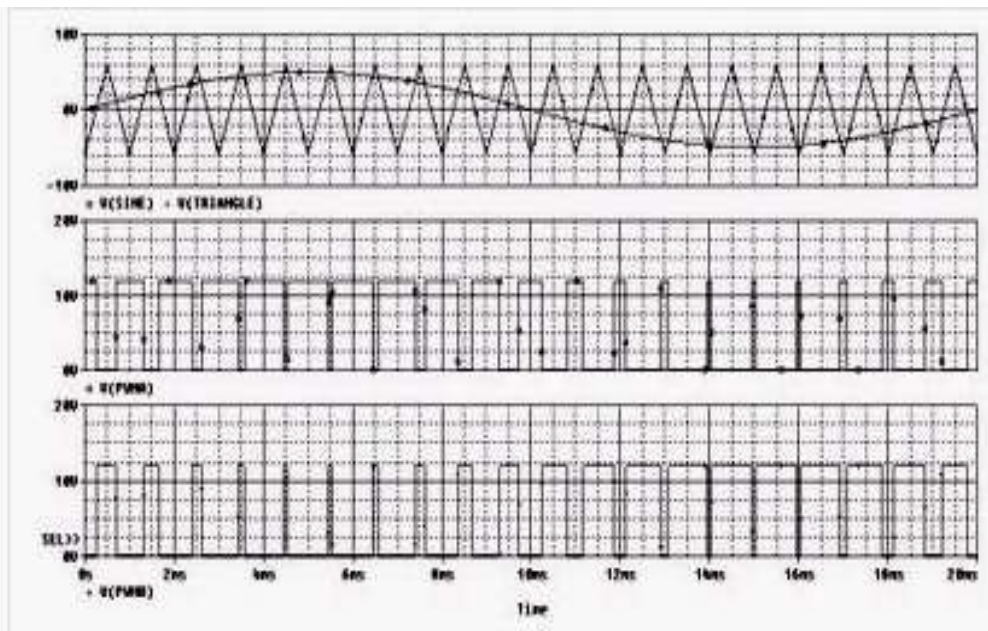


Figure 4.3

2. Uni-polar PWM

This method of PWM allows transitions in the output between V and zero and V_0 and

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zero. The Figure 4.4 clearly shows how a uni-polar PWM signal is generated. The voltages specified on this diagram are with reference to the full bridge circuit.

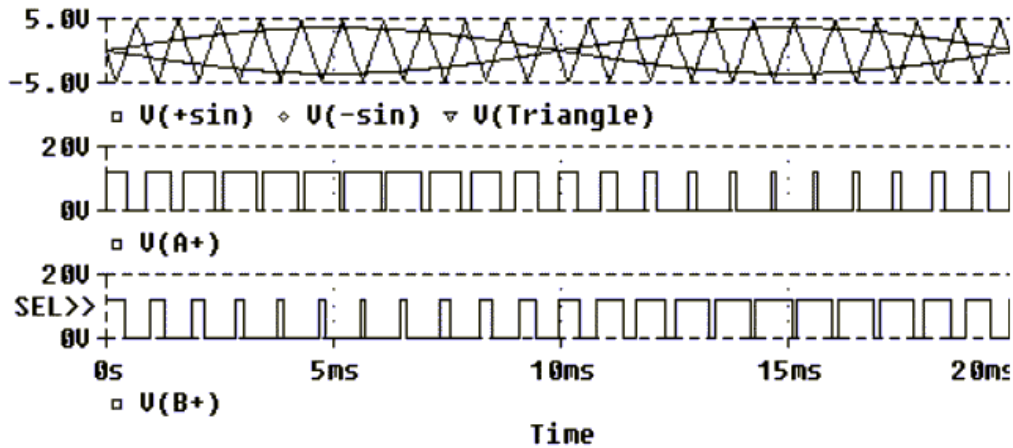


Figure 4.4

The main reason that uni-polar switching is preferred over bi-polar switching is that it effectively doubles the switching frequency and at the same time cuts maximum voltage transitions in half. A simple analysis of figure 4.4 shows that the both output voltages are 90 degree out of phase with each other. The switching frequency harmonic components of both are however have the same phase. Thus, switching frequency harmonics disappear from the output voltage. The sidebands of the switching frequency harmonic component also disappear, as does the dominant harmonic at twice the switching frequency. Clearly then, uni-polar switching provides a significant reduction in EMI.

PWM SOFTWARE IMPLEMENTATION:

The pattern at which the duty cycle of a PWM signal varies can be created through simple analog components, a digital microcontroller, or specific PWM integrated circuits. We go for micro controller and for this we have selected PIC16F877A. This

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controller has two CCP (PWM mode) module and 7 analog to digital converter module that are enough for our requirement.

STEPS FOR PROGRAMMING PIC16F877A FOR PWM OPERATION:

The following steps should be taken when configuring the CCP module for PWM operation:

1. Set the PWM period by writing to the PR2 register.
2. Set the PWM duty cycle by writing to the CCPR2L register and CCP2CON<5:4> bits.
3. Make the CCP2 pin an output by clearing the TRISC<1> bit.
4. Set the TMR2 pre-scale value and enable Timer2 by writing to T2CON.
5. Configure the CCP2 module for PWM operation.

SOFTWARE FLOW FOR SINE WAVE CREATION:

The software detail that we have discussed up till now can be used to create a square wave with fix period and duty cycle. For our case we need square wave with varying duty cycle. Since we know that duty cycle is control by CCP2H register and Period is control by PR2 register and since we need to vary duty cycle only and will keep period of pulses constant so CCP2H is the only register that we need to pay attention at. To vary PWM duty cycle we need to load CCP2H with new value each time the TMR2 register is match with PR2 register. Now question is that how often we need to do that and with what value we should load it. The answer to these critical questions is found in basic Digital Signal Processing (DSP).

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The formula for the digitized sine wave obtained by sampling a sine wave can be written as

$$\text{Sine}(n) = 5 \times \sin(n \times 180/N) \text{-----(1)}$$

Where

N= total number of sample

n= index for sample

we have used 5 assuming that maximum allowable level for logic high is 5 and 180 because we are only working for half cycle.

In our case since we need to get 50hz at output so taking it half is equal to 100hz resulting in the time period of 10 millisecond for half wave. Now if we take each pulse of 250 micro second then we require 40 such pulses to come upto the total of 10 msec. thus time period of CCP2 module is set to 250 usec. This is done by loading PR2 register with 249. Now equation 1 can be written as

$$\text{Sine}(n) = \text{PR2} \times \sin(n \times 180/40) \text{-----(2)}$$

This result is in a table 4.1 which is loaded one by one in CCP2H on each compare and match.

Indexes(n)	Sine(n)
1	0
2	20
3	39
4	58
5	77
6	95
7	113
8	130
9	146
10	162

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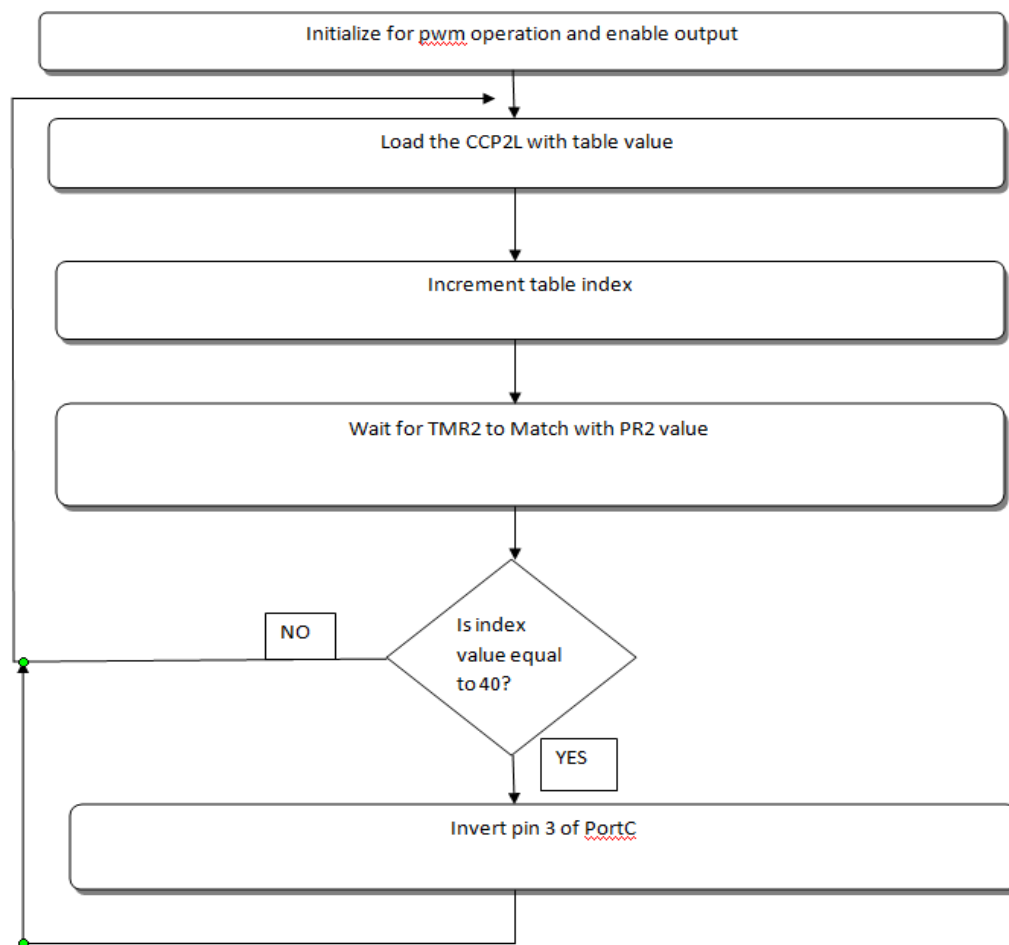
11	176
12	189
13	201
14	212
15	222
16	230
17	237
18	242
19	246
20	248
21	249
22	248
23	246
24	242
25	237
26	230
27	222
28	212
29	201
30	189
31	176
32	162
33	146
34	130
35	113
36	95
37	77
38	58
39	39
40	20

Table 4.1

Since this is for half cycle of sine wave so what about the other half....?

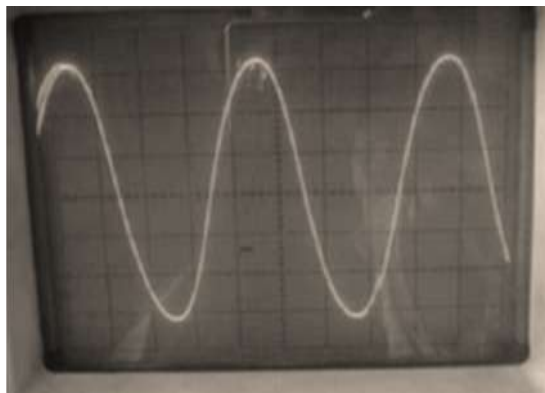
As both half parts of sine wave are essentially identical apart from one is flip. So all we need to do is to flip the output after every 10 msec. This is achieve by hardware (please refer to full bridge driver part in hardware section). In software all we need to do is to invert on pin after every 10 msec. For this we have assigned pin 3 of Port C and to enable the output we have assigned pin 4 of Port C. So the whole flow of software is:

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SECTION: 4**RESULTS:**

After putting our design to work, we were able to make a 60 watt high frequency charger which can deliver up to 5 ampere. The second part of our project i.e. inverter. We were able to make a 500 watt small size, light weight sine wave inverter and after the careful handling of all the control signals. We obtained a 50 HZ Sine wave at the output as shown below:



RECOMMENDATIONS AND FUTURE WORK:

Although almost all of the goals were met in the project but there are several features that can be improved upon:

- Filter at the output which extracts the sine wave can be design of higher order in order to improve its quality and decreases the distortion.
- Control section can be made more intelligent. Load regulation and monitoring of other parameters can be improved because there can be loads which can cause voltage fluctuations and spark.
- Transformer and inductor can be design more efficiently, in order to reduce all types of losses and hence increasing output power

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